

• By Theorem 12, T is one-to-one.

• From Theorem 12, we know T maps $\mathbb{R}^2$ onto $\mathbb{R}^3$ if and only if the columns of A span $\mathbb{R}^3$.

• by defn, if the vector $a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \in \text{Span} \left\{ \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 7 \\ 3 \end{bmatrix} \right\}$, $\exists c_1, c_2 \in \mathbb{R}$.

s.t. $c_1 \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 7 \\ 3 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$

• From Theorem 3, the above equation has the same solution set as the system of equations corresponding to the augmented matrix

$$\begin{bmatrix} 3 & 1 & a_1 \\ 5 & 7 & a_2 \\ 1 & 3 & a_3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 & a_1 \\ 5 & 7 & a_2 \\ 1 & 3 & a_3 \end{bmatrix} \xrightarrow[\text{Row 1 and Row 3}]{\text{Interchange}} \begin{bmatrix} 1 & 3 & a_3 \\ 5 & 7 & a_2 \\ 3 & 1 & a_1 \end{bmatrix} \xrightarrow[\text{-5xRow 1}]{\text{Row 2 = Row 2}} \begin{bmatrix} 1 & 3 & a_3 \\ 0 & -8 & a_2 - 5a_3 \\ 3 & 1 & a_1 \end{bmatrix}$$

$$\xrightarrow[\text{3xRow 1}]{\text{Row 3 = Row 3 -}} \begin{bmatrix} 1 & 3 & a_3 \\ 0 & -8 & a_2 - 5a_3 \\ 0 & 1 & a_1 - 3a_3 \end{bmatrix} \xrightarrow[\text{x(-1/8)}]{\text{Row 2 = Row 2}} \begin{bmatrix} 1 & 3 & a_3 \\ 0 & 1 & (5a_3 - a_2)/8 \\ 0 & 1 & a_1 - 3a_3 \end{bmatrix} \xrightarrow[\text{-Row 2}]{\text{Row 3 = Row 3 -}} \begin{bmatrix} 1 & 3 & a_3 \\ 0 & 1 & (5a_3 - a_2)/8 \\ 0 & 0 & a_1 + a_2/8 - 19a_3/8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & a_3 \\ 0 & 1 & (5a_3 - a_2)/8 \\ 0 & 0 & a_1 + a_2/8 - 19a_3/8 \end{bmatrix}$$

Verification of whether the final matrix is in echelon form

1. If $a_1 + a_2/8 - 19a_3/8 \neq 0$, then each of the rows are non-zero.

$\therefore$ The condition that all the non-zero rows must be above the zero rows is vacuously True.

If $a_1 + a_2/8 - 19a_3/8 = 0$, then Row 1 and Row 2 are non-zero rows. Row 3 is a zero row. $\therefore$ The condition that all the non-zero rows must be above the zero rows is True.

2. Row 1 Leading entry = 1 in Column 1

Row 2 Leading entry = 1 in Column 2

Row 3 Leading entry = $a_1 + a_2/8 - 19a_3/8$ in Column 3.

$\therefore$ The condition that for each row, the leading entry must be in a